

ADITYA CHUNDURI

(678) 776-0805 | chunduri@usc.edu | linkedin.com/in/aditya-chunduri | github.com/Chunduri-Aditya
chunduri-aditya.github.io/Portfolio

Education

University of Southern California

Master of Science in Applied Data Science

Los Angeles, CA

January 2024 – December 2025

SRM Institute of Science and Technology

Bachelor of Technology in Computer Science and Engineering (AI/ML Specialization)

Chennai, India

June 2019 – July 2023

Technical Skills

Languages: Python, C++, SQL, JavaScript, Bash

LLM and AI: Inspect AI, AgentDojo, LangChain, Hugging Face, Ollama, ChromaDB, RAG, DPO, adversarial evaluation, prompt injection, red teaming

MLOps and Cloud: Docker, AWS, FastAPI, Flask, Git, GitHub Actions, CI/CD, Gradio

Data and Tools: PostgreSQL, Neo4j, Pandas, NumPy, Plotly, PyTorch, TensorFlow, Scikit-learn

Experience

USC – Viterbi School of Engineering

Los Angeles, CA

Research Assistant, Computer Vision and Medical Imaging

August 2024 – December 2024

- Designed structured model evaluation methodology across 3 concurrent research projects, implementing experiment tracking to ensure reproducible, comparable results across architectures.
- Built an automated data pipeline with quality-validation controls processing 10,000+ medical images, cutting manual curation time by 40%.
- Presented model performance findings in weekly cross-functional reviews, translating quantitative results into actionable research decisions.

SSN College of Engineering

Remote

Research Intern, Computer Vision and Object Detection

June 2021 – July 2021

- Automated a video dataset generation pipeline, eliminating 100% of manual annotation overhead across frame-level labeling workflows.
- Produced 10+ reusable deep learning modules in PyTorch and TensorFlow, standardizing team workflows and reducing onboarding time for new contributors.

Projects

Agent Shield: LLM Agent Security Evaluation Framework | Python, Inspect AI, AgentDojo

April 2026 – Present

- Designing an 8-module adversarial evaluation framework stress-testing LLM agents for prompt injection, MCP tool poisoning, RAG memory attacks, multi-agent manipulation, covert exfiltration, and Cialdini-grounded behavioral drift; built on Inspect AI (UK AISI harness).
- Benchmarking across 8 frontier models (Claude, GPT-4.1, Gemini 2.5, Llama 3.1, Mistral Large, DeepSeek V3) with unified ASR and utility-under-attack metrics; arxiv preprint in progress.

AI RemixMate: AI-Powered DJ Mixing Engine | Python, PyTorch, Librosa, Demucs, Gradio

September – December 2025

- Built an end-to-end audio ML pipeline orchestrating Demucs source separation, Librosa spectral analysis, and Camelot-Wheel harmonic matching; deployed via FastAPI with GPU acceleration.
- Shipped Gradio web interface handling 8+ audio formats with real-time BPM, key, and energy analysis; open-sourced with 9 GitHub stars.

AkashicTree: Automated Content Generation Pipeline | Python, Diffusers, Ollama, ElevenLabs, FLUX.1

June – August 2025

- Architected a model-agnostic LLM orchestration layer supporting drop-in substitution between local Ollama models and cloud APIs (GPT-4o-mini), reducing inference costs by 90% during development cycles.
- Built a unified inference abstraction producing text, image, and audio from a single prompt brief.

AI Health Journal: Privacy-First LLM Assistant | Python, Flask, Ollama, ChromaDB

January – May 2025

- Independently built a production RAG (Retrieval-Augmented Generation) system: ChromaDB vector store, Ollama 7B-parameter local inference, Flask API layer; zero external API dependencies.
- Applied DPO (Direct Preference Optimization) fine-tuning for domain-specific health response alignment; optimized vector retrieval achieving sub-second context injection across 10,000+ document chunks.

Model-Behavior-Lab: Adversarial LLM Evaluation Platform | Python, Ollama, Plotly, CI/CD

September – December 2024

- Built an adversarial LLM evaluation platform running 5,000+ automated tests across 10+ frontier models (GPT-4, Claude, Llama, Mistral), measuring instruction-hierarchy violations, hallucination rate, and refusal consistency with zero manual review.
- Integrated CI/CD so new model versions benchmark automatically; structured JSON test suite supports any Ollama or API-accessible model as a drop-in target for attack evaluation.

Publication

Wind Power Analysis Using Machine Learning – IJRASET, August 2023 (Co-Author): Co-developed a neural network framework integrating numerical weather data for wind energy forecasting across multiple datasets.